

B. Tech.Degree III Semester Examination November 2012

IT/CS/EC/ME/EB/EI 302 ELECTRICAL TECHNOLOGY (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 x 5 = 40)

- I. (a) Classify the two different types of losses occurring in a transformer. Discuss the methods to minimize these losses.
- (b) Draw and explain the phasor diagram of practical transformer when it is connected to an inductive load.
- (c) List any four conditions for voltage build up in dc shunt generator.
- (d) Explain the significance of back emf in dc motors.
- (e) Derive the emf equation of an alternator.
- (f) Explain the working of capacitor start single phase induction motor with relevant diagrams.
- (g) What is meant by corona? List any three factors which affects corona.
- (h) List any five advantages of DC transmission over AC transmission.

PART B

(4 x 15 = 60)

- II. (a) Derive an expression for saving of copper in an auto transformer as compared to an equivalent two winding transformer. (5)
- (b) A 11000/230V, 150 KVA, Single phase 50 Hz transformer has core loss of 1.4kw and full load copper loss 1.6 kw. Determine (i) the KVA load for maximum efficiency and the value of maximum efficiency at unity power factor (ii) the efficiency at half full load, 0.8 p.f. leading. (10)
- OR**
- III. (a) Explain how current transformers and potential transformers are used as instrument transformers. (5)
- (b) A 10 KVA, 200/400V, 50 Hz, 1 ϕ transformer gave the following test results: (10)
OC – 200V, 1.3A, 120W
SC – 22V, 25A, 200W
Draw the equivalent circuit referred to L.V. side.
- IV. (a) Classify D.C. generators based on their excitation. (5)
- (b) A shunt generator supplies 96A at a terminal voltage of 200V. The armature and shunt field resistance are of 0.1 Ω and 50 Ω respectively. The iron and frictional losses are 2500w. Find: (i) emf generated (ii) power developed in armature (iii) copper losses (iv) commercial efficiency. (10)
- OR**
- V. (a) Discuss any two speed control methods of DC shunt motor. (5)
- (b) Sketch the mechanical and electrical characteristics of DC series motor and explain why it is not started under no load condition. (5)
- (c) Explain the necessity of starters in DC motors. (5)

(P.T.O)

- VI. (a) Explain emf method for finding out regulation of an alternator. (5)
- (b) A 3 phase, star connected alternator is rated at 1600 KVA, 13500V. It has an armature resistance of 1.5Ω per phase. A field current of 40A produces a short circuit current of 200A and an open circuit emf of 1100V (line to line). Calculate the percentage voltage regulation for a load of 1280kw at: (i) 0.8p.f. leading (ii) 0.8 p.f. lagging. (10)

OR

- VII. (a) Synchronous motors are not self starting. Justify the statement. (5)
- (b) Explain star-delta starter used for starting a 3ϕ induction motor. (5)
- (c) Derive the condition for maximum running torque of a 3ϕ induction motor. (5)

- VIII. (a) With neat schematic diagram explain the working of hydroelectric power plants. (12)
- (b) Describe skin effect. (3)

OR

- IX. (a) Explain load dispatching. (5)
- (b) Discuss on any five electrical equipments in a substation. (10)